

Builder Basics — The Trail Maker (Extension · Intermediate)

Topic: Place On Move · **Course:** Builder Basics · **Type:** Extension Activity · **Level:** Intermediate · **Time:** about 38 minutes

✖ This is a bonus challenge, not a weekly lesson. Try it once you are comfortable placing blocks and you want the agent to place a whole line for you.

In Week 2 you placed blocks one at a time. This world has a faster way: type `run` and the agent flips a switch called **place on move**, then walks forward. Understand the **idea** — you won't type any code.

When **place on move** is **ON**, the agent drops a block on every step — a **trail**. When it is **OFF**, it walks but places nothing — a **gap**. The program flips ON, OFF, then ON again, so the trail comes out with a gap in the middle.

1 · Predict 🧠

Here is the order the program runs:

```
place on move ON
move forward by 4
place on move OFF
move forward by 2
place on move ON
move forward by 4
```

Each move places one block per step while the switch is ON. How many blocks come from the first `move forward 4`? How many from the `move forward 2` while OFF? How many from the last `move forward 4`? Write all three numbers.

**Add them up. How many blocks are placed in total, and how wide is the gap in the middle?
Show your working.**

2 · Trace the Switch

Walk through the program one move at a time. For each move command, write the switch state and how many blocks that stretch places.

Move command	Switch (ON / OFF)	Blocks placed
move forward 4		
move forward 2		
move forward 4		

The agent walked 10 steps but placed fewer than 10 blocks. Why? Which stretch is the reason?

3 · Reason About the Switch

No code here — think it through.

The trail comes out as one solid line with NO gap. The switch must have been left in the wrong state somewhere. Which line of the program went wrong for that to happen? Explain in one or two sentences.

Now imagine the very first **place on move ON** was missing, so the switch stayed OFF the whole time. What would be on the ground after the agent finished walking? Explain why.

4 · Make It Yours

The switch decides WHEN the agent places, and the move numbers decide HOW MANY blocks and how wide the gap is.

Design a trail that places exactly 10 blocks with a 3-wide gap in the middle. Write your ON / OFF / move sequence, line by line.

One sentence: how did you pick your move numbers to get 10 blocks?

5 · Show Your Work 📷 🎥

Open the **Trail Maker** world. Type **run** in the chat to watch the agent lay a trail with a gap. Then try your own version from Section 4 — change the move numbers and the ON / OFF switches, and run it again.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.